

JUNSEN LI

Objective: AI Applications / LLM Agent Engineering (Internship)

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Nanjing University · School of Engineering Management · Computer Science & Technology (CS-Finance Experimental Class), Sophomore

Available full-time during vacations (6 days/week) | Remote during term | Nanjing preferred

Born Feb 2007 | github.com/jason218-ljs



EDUCATION

Nanjing University — School of Engineering Management, Computer Science & Technology (CS-Finance Experimental Class)

Sep 2025 – Jun 2029 (exp.)

- GPA **4.28 / 5.0**; CET-4: **579** (CET-6 scheduled Dec 2026)
- Core courses: Data Structures & Algorithms, Computer Systems Fundamentals, Discrete Mathematics, Machine Learning, Databases, Probability & Statistics

INTERNSHIP EXPERIENCE

Transwarp Technology — AI-Native Portfolio Program (University-Industry Internship), Quantitative Researcher

Jul – Aug 2026

AI-native portfolio investment program · Member of a 5-person Investment Committee with risk veto; owned factor scoring, portfolio construction, backtest validation, risk checks, and attribution review

- Led development of AlphaForge**, a multi-factor stock-screening and backtesting system: designed a 100-point three-layer scoring framework (fundamental / technical / money-flow) with adaptive thresholds across 21 sub-industries, a two-stage “pattern pre-filter + quality scoring” pipeline, 214 automated tests and a web dashboard
- Built an event-driven backtesting engine** modeling A-share market frictions (T+1, price limits, stamp duty, slippage) and Kelly-criterion position sizing with per-stock and per-industry risk constraints, validated on real A-share data
- Led design of a three-flywheel self-evolution system** (engineering / data / cognition): a zero-intrusion sidecar architecture for health checks, data-quality detection and methodology review. In production it detected a 100%-missing data field and clustered 139 monitoring anomalies into 9 error patterns; backed by 7 zero-intrusion red lines, 62 unit tests, CI and static rule scanning
- Contributed to Buffett-Value-Agent and a Human-AI Boundary Protocol (HABP)**: defined LLM-agent output boundaries through a triple-challenge mechanism (data provenance → assumption identification → reverse logical deduction), and implemented rule-based modules for deterministic financial computation and risk validation

PATENT & RESEARCH

Invention Patent — “Method and System for Trustworthy Uncertainty Constraint of AI Agents”

3rd Inventor · Filed 2026

- Trustworthy-control technology for LLM-based agents, constraining behavioral, semantic and provenance uncertainty in multi-step execution

Yuxiu College “Young Scientist Growth Camp” — Early Research Training: Social-Media Investor Sentiment

Mar – Jun 2026

- Lead crawler developer**: collected Xueqiu data in three batches with Requests and browser automation, handling dynamic loading and anti-scraping, plus deduplication, cleaning and data-quality documentation
- Contributed to cross-platform investor-sentiment and market-attention research using LLM-assisted annotation and Python data processing; annotated A-share forum posts and distilled the annotation guideline

PROJECTS

2026 CUMCM Problem B — Rapid Automatic Localization and Clearance of Radio Interference Sources

Team Captain · Modeler · Programmer | Aug – Sep 2026

- Coordinated a 3-person team and led end-to-end modeling and implementation across all four sub-problems, delivering **7,131 lines** of solver code (1,683 for Problem 1; 1,661 for Problem 2); owned all figures and appendix code organization
- Established a **coverage-guarantee theorem** reducing “100% clearance” to an independently provable geometric-coverage problem, and exposed via a constructive counterexample that a diameter-based circle cannot cover the localization region (seen in **8.08%** of 20,000 randomized intersections)
- Designed a “center + 9-point ring” guaranteed station layout with fused online path planning: 200 runs plus a 600-run revalidation both achieved **100% clearance at 270.8 s/source** (lower bound ≈ 229 s). For direction-unknown sources, an inner-outer ring encirclement drove the expected miss rate to strictly **zero**: 120 runs, 100% clearance, **542.9 s/source, only 13% above the 480 s lower bound**

Personal Bilingual Website · Portfolio | Next.js · TypeScript · AI-assisted development

Sep 2026

- Independently built a bilingual document system, automatic content indexing, build-time content validation and cloud deployment; maintained as a personal portfolio

MEM Course Project, “Engineering Opportunity Compass — Market-Opportunity Intelligence for Engineering Firms” · AI Development Lead

7-member team | Sep 2026 – Present

- Owned the rag-kb knowledge-base skill (init / ingest / query), the opportunity-extraction pipeline and the scoring engine; stack: Bailian Embedding + Rerank + Milvus Lite

SKILLS

- Languages**: Python (proficient — modeling, data processing, engineering), C / C++, SQL (basic), TypeScript (basic)
- AI & ML**: LLM-agent applications & prompt engineering, Human-AI Boundary Protocol (HABP) design, machine learning fundamentals (coursework + self-study)
- Data & Engineering**: pandas / NumPy / matplotlib, pytest, Git, GitHub Actions CI, Linux / Docker (basic)
- Algorithms**: LeetCode 100+ (ongoing) | graph & shortest-path algorithms, linear / integer / nonlinear programming, clustering